INCH-POUND

MIL-DTL-28837/9B
30 July 2015
SUPERSEDING
MIL-DTL-28837/9A
25 October 2006

DETAIL SPECIFICATION SHEET

MIXER STAGES, RADIO FREQUENCY, DOUBLE BALANCED, TO CONFIGURATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28837.

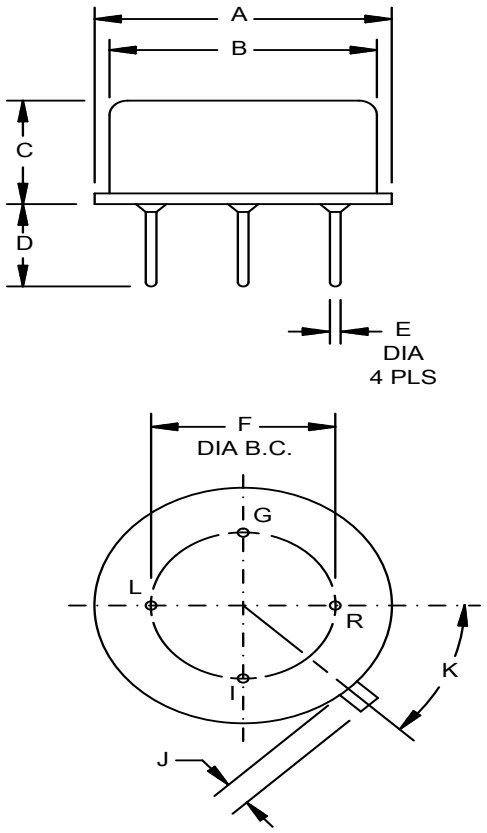


FIGURE 1. Outline drawing for mixers PIN M28837/9-01 through M28837/9-04.

AMSC N/A

FSC 5895

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Ltr	Inches		millimeters	
	Max	Min	Max	Min
A	.510	.498	12.95	12.65
B	.455	.445	11.56	11.30
C	.220	---	5.59	---
D	---	.175	---	4.44
E	.020	.016	0.51	0.41
F	.310 DIA B.C.	.290 DIA B.C.	7.87 DIA B.C.	7.37 DIA B.C.
J	.050	---	1.3	---
K	48° Ref	42° Ref	48° Ref	42° Ref

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (in millimeters) are given for general information only.
3. Dimensions are in accordance with ASME Y14.5M.

FIGURE 1. Outline drawing for mixers PIN M28837/9-01 through M28837/9-04 - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Electrical characteristics:

Operating frequency range: See table I.

LO drive power: See table I.

Conversion loss (max): See table I.

Noise figure (SSB): See table I.

Isolation (minimum): See table I.

Maximum input power: See table I.

Conversion compression: See table I.

Desensitization (max): See table I.

DC relative polarity: See table I.

VSWR: See table I.

Third order two tone intermodulation: See table I.

Impedance: 50 ohms.

Physical and environmental characteristics:

Weight: .07 ounce (2 grams) maximum.

Temperature:

Operating: -54°C to +100°C.

Non-operating: -65°C to +100°C.

Shock: Test condition C, MIL-STD-202.

Life: 100,000 hours minimum.

Seal: Hermetic.

Part or Identifying Number (PIN): M28837/9 (dash number from table I).

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TABLE I. Electrical characteristics.

Dash no.	Operating frequency range (MHz)	LO input drive power in dBm 1/	Maximum conversion loss (SSB) dB (MHz)	Noise figure (SSB) dB (MHz)	Isolation Minimum LO-RF dB	Isolation Minimum LO-IF dB	Isolation Minimum RF-IF dB	Isolation Frequency (MHz)
01N 01S	RF 10-1,000 LO 10-1,000 IF DC-1,200	minimum -3 TV 0 maximum +5	8.5 at f_L and f_R 10-1,000 f_I DC-1,200	Within 1 dB of conversion loss	40	35	30	10-50
					20	25	15	50-750
					15	17	10	750-1,000
02N 02S	RF 10-1,000 LO 10-1,000 IF DC-600	Minimum +17 TV +20 maximum +24	7.5 at f_L and f_R 10-300 f_I 10-200	7.5 at f_L and f_R 10-300 f_I 10-200	35	35		10-100
					25	25		100-400
			9.0 at f_L and f_R 10-300 f_I 1-600	9.0 at f_L and f_R 10-300 f_I 2-600	18	18		400-1,000
			10 at f_L and f_R 10-1,000 f_I 1-500	10 at f_L and f_R 10-1,000 f_I 2-500			12	10-1,000
03N 03S	RF 10-1,500 LO 10-1,500 IF DC-800	minimum +7 TV +7 maximum +13	7.2 at f_R 20-600 f_L 10-800 f_I 1-200	7.2 at f_R 20-600 f_L 10-800 f_I 1-200	35	30		10-500
					28	20		500-1200
			9.0 at f_L and f_R 10-1,500 f_I 1-800	9.0 at f_L and f_R 10-1,500 f_I 1-800	25	18		1200-1500

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash no.	RF and LO 2/ maximum power input (rms)	Conversion compression (maximum)	Desensitization (maximum)	Relative dc polarity 3/	VSWR (maximum)			Third order, two tone intermodulation	
					LO	IF	RF	dBm	Frequencies
01N 01S	200 mW	1.0 dB @ P_R -8 dBm	1.0 dB @ P_{R2} -10 dBm	Positive	2.0:1	---	2.0:1	+7 4/	f_{LO}: 600 MHz f_{R1}: 500 MHz @ - 20 dBm f_{R2}: 505 MHz @ -20 dBm
02N 02S	500 mW	1.0 dB @ P_R +20 dBm	1.0 dB @ P_{R2} +18 dBm	Negative	2.5:1	2.0:1	3.5:1	+25 4/ +23	f_{LO}: 15 MHz @ +23 dBm f_{R1}: 10 MHz @ 0 dBm f_{R2}: 11 MHz @ 0 dBm f_{LO}: 400 MHz @ +23 dBm f_{R1}: 1000 MHz @ 0 dBm f_{R2}: 900 MHz @ 0 dBm
03N 03S	200 mW	1.0 dB @ P_R 0 dBm	1.0 dB @ P_{R2} -2 dBm	Positive	2.1:1	2.8:1	3.0:1	+12 4/ +10	f_{LO}: 15 MHz f_{R1}: 10 MHz @ -10 dBm f_{R2}: 11 MHz @ -10 dBm f_{LO}: 1500 MHz f_{R1}: 1400 MHz @ -10 dBm f_{R2}: 1410 MHz @ -10 dBm

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash no.	Operating frequency range (MHz)	LO input drive power in dBm 1/	Maximum conversion loss (SSB) dB (MHz)	Noise figure (SSB) dB (MHz)	Isolation Minimum LO-RF dB	Isolation Minimum LO-IF dB	Isolation Minimum RF-IF dB	Isolation Frequency (MHz)
04N 04S	RF 10-1,600 LO 10-1,600 IF DC-800	Minimum +11 TV +13 Maximum +17	7.5 at f_L 10-800 f_R 20-600 f_I 0.4-200	7.5 at f_L 10-800 f_R 20-600 f_I 0.4-200	35	28		10-500
					28	20		500-1,000
			8.5 at f_L and f_R 10-1,600 f_I 0.4-800	8.5 at f_L and f_R 10-1,600 f_I 0.4-800	25	15		1000-1,600

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash no.	RF and LO 2/ maximum power input (rms)	Conversion compression (maximum)	Desensitization (maximum)	Relative dc polarity 3/	VSWR (maximum)			Third order, two tone intermodulation	
					LO	IF	RF	dBm	Frequencies
04N 04S	200 mW	1.0 dB @ P_R +7 dBm	1.0 dB @ P_{R2} +5 dBm	Positive	2.8:1	3.0:1	3.0:1	+18	f_{LO}: 1,150 MHz f_{R1}: 500 MHz @ -5 dBm f_{R2}: 505 MHz @ -5 dBm

1/ Unless otherwise specified, TV is the test value of the LO power for the electrical characteristics.

2/ These values are for +25°C and are derated linearly to +100°C.

3/ With two in-phase signals applied to the LO and RF ports.

4/ Input intercept points.

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Referenced documents. In addition to MIL-DTL-28837, this document references the following:
MIL-STD-202
ASME Y14.5M

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Custodians:
Army - CR
Navy - EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5895-2015-005)

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS
Air Force -19, 99

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